Hyperspectral Image Denoising With Weighted Nonlocal Low-Rank Model and Adaptive Total Variation Regularization

Yang Chen, Wenfei Cao, Li Pang, and Xiangyong Cao

***Abstract*— Hyperspectral image (HSI) is always corrupted by various types of noises during image capturing, such as Gaussian noise, stripe noise, deadline noise, impulse noise, and more. Such complicated noise significantly degrades imaging quality and thus limits the performance of downstream vision tasks. Current HSI denoising methods tackle this problem by modeling either the spectral–spatial prior of HSI or the noise characteristic of HSI, and few works consider the two aspects simultaneously. In this article, we propose a new HSI denoising method by simultaneously modeling the HSI prior and the HSI noise characteristic. Specifically, we first utilize the nonindependent and identically distributed (non-i.i.d.) mixture of Gaussian (MoG) assumptions to characterize the complex noise, which corresponds to optimizing a weighted fidelity function. Second, we exploit HSI's nonlocal similarity and spatial–spectral correlation priors by applying a nonlocal low-rank model. Third, we design an adaptive edge-preserving total variation (TV) regularization term to characterize the nonlocal smooth property of HSI. Finally, we propose a new denoising model and develop an effective alternating direction method of multipliers (ADMM) algorithm to solve it. Extensive experiments on simulated data and real data substantiate the superiority of the proposed method beyond state-of-the-art.**

***Index Terms*— Adaptive spatial–spectral total variation (ASSTV), hyperspectral image (HSI) denoising, nonindependent and identically distributed (non-i.i.d.) noise modeling, nonlocal low-rank model.**

I. Introduction

HYPERSPECTRAL image (HSI) is a high-dimensional data cube containing an array of 2-D images collected from sensors over various bands and provides rich spectral information, and thus has been widely used in military surveillance, terrain detection, mineral exploration, pharmaceutical counterfeiting, and food safety [1], [2], [3]. However, due to the sensor sensitivity, calibration error, and physical mechanism, HSI is unavoidably contaminated by various kinds of noise, such as Gaussian noise, stripe noise, deadline noise, impulse noise, and so on [4], [5], [6]. This noise severely degrades the quality of the imagery and limits the performance of the subsequent processing, that is, classification [7], unmixing [8], [9], and target detection [10]. Therefore, noise removal is a critical research topic in the HSI processing area.

The simplest way for an HSI denoising task is to directly apply the traditional 1-D or 2-D denoising methods pixel by pixel [11] or band by band [12], [13], [14], [15], [16]. However, these methods usually could not provide high-quality results since they ignore the correlations among spectral bands or adjacent pixels. In recent years, many advanced HSI denoising methods have been proposed, among which the works exploring and modeling the intrinsic prior knowledge of the HSI have made the most outstanding achievements. The most important prior knowledge of HSI mainly includes spatial and spectral correlation, nonlocal spatial similarity, and nonlocal smoothness. Specifically, spectral correlation is based on the fact that the images in adjacent bands are usually collected with similar sensor parameter settings and thus have a high correlation. Therefore, the HSI should lie in a low-dimensional spectral subspace. By reshaping the original 3-D HSI data cube into a 2-D matrix, the low-rank matrix analysis methods can be directly used to recover the intrinsic spectral subspace [17], [18], [19], [20], [21], [22]. The nonlocal spatial similarity prior is based on the fact that there are always similar patterns in image spatial space, and thus the low-rank-based filtering methods can be applied to remove the noise from the grouped similar patterns [23], [24], [25], [26]. The local smoothness prior is based on the fact that similar objects are often distributed in a local area with high probability. Besides, the adjacent bands of images are usually collected with similar sensor parameter settings, which results in similar values. Such local smooth prior can be encoded by using the total variation (TV) technique on the spatial and spectral domains of the HSI [27]. Additionally, more advanced TV methods are proposed to characterize this prior [28], [29], [30], [31].

To further improve the denoising performance, current state-of-the-art HSI denoising methods mainly consider the combination of the aforementioned three types of priors [32], [33], [34]. For example, the LRTV method [35] and SAWTV

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Yang Chen and Wenfei Cao are with the School of Mathematics and Statistics, Shaanxi Normal University, Xi'an 710119, China (e-mail: yangchen9103@163.com; caowenf2015@gmail.com).

Li Pang is with the School of Electronic and Information Engineering, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: 2195112306@stu.xjtu.edu.cn).

Xiangyong Cao is with the School of Computer Science and Technology, Xi'an Jiaotong University, Xi'an 710049, China, and also with the Ministry of Education Key Laboratory for Intelligent Networks and Network Security, Xi'an 710049, China (e-mail: caoxiangyong@mail.xjtu.edu.cn).

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method [36] take the spectral correlation and nonlocal smoothness into consideration. The NGmeet method [37] and its derivatives [38], [39] consider the spectral correlation and nonlocal similarity by mapping the original HSI into the low-dimensional spectral subspace first and then denoising the mapped image. Additionally, the LLRT method [40] and its derivatives [41] combine the nonlocal self-similarity with nonlocal smoothness by utilizing the unidirectional low-rank tensor recovery model with the TV regularization. The low-rank tensor decomposition-based methods [42], [43], [44] consider both the spatial and spectral correlations. By extra considering the nonlocal smoothness prior, the spatial–spectral TV regularization was integrated into the low-rank tensor model [45], [46], [47]. To further utilize the nonlocal similarity prior, the nonlocal tensor decomposition models with TV regularization were proposed in [48], [49], [50], and [51].

Recently, learning-based methods have been gradually applied to HSI denoising, by training with large amounts of data to learn the deep HSI prior. For instance, Chang et al. [52] first introduced the deep convolutional neural network (CNN) to extra spatial and spectral information. Toward bandwise HSI denoising, Yuan et al. [53] developed a deep residual CNN with multiscale and multilevel feature representation. Zhang et al. [54] presented a deep CNN by incorporating the spatial–spectral gradient information. Zhang et al. [55] proposed a spatial–spectral Bayesian posterior (DSSBP) model and solved by deep network. To learn the spatial and spectral correlation better, Shi et al. [56] proposed a 3-D attention network with two separate branches. Pan et al. [57] proposed quasiattention recurrent networks to explore the relevance of spatial features among the spectral domain. Dong et al. [58] presented a modified 3-D U-net architecture to exploit spatial–spectral correlations. Wei et al. [59] introduced a quasirecurrent pooling function to further capture the global correlation along the spectrum, and Cao et al. [60] proposed a global reason network to encode the nonlocal spatial information and global spatial–spectral correlations with three well-designed modules.

All of the aforementioned denoising methods based on encoding multipriors have obtained good performance in removing one or two types of noises. However, in real scenarios, the noise in HSI is much more complicated and does not have independent identically distributed statistical characteristics, while always presenting the nonindependent and identically distributed (non-i.i.d.) structure [22]. The bias between the real noise distribution characters and the noise assumption of the model greatly limits the performance of these methods. To alleviate this issue, a noise modeling methodology has been proposed. Meng and De la Torre [61] first modeled the HSI noise with an i.i.d. mixture of Gaussian (MoG) distribution since the MoG constructs a universal approximator to any continuous density function in theory. Later, Cao et al. [21] and [62] proposed a more general i.i.d. mixture of exponential power (MoEP) distribution to model the complex HSI noise. Besides, based on the mixture noise assumption, He et al. [20] proposed a noise-adjusted low-rank method and Wang et al. [63] proposed a GLRR denoising method for the reconstruction of the corrupted HSIs. Furthermore, Barron [64] proposed a more general noise model which derived the general loss function. Chen et al. [22] observed that the complex HSI noise always exhibits non-i.i.d. statistical characteristic and thus proposed a non-i.i.d. MoG (NMoG) to characterize the noise, which significantly improves the denoising performance of complex noise scenario and has been accepted as a reasonable noise hypothesis.

In this article, we propose a new HSI denoising method that combines the noise modeling methodology with the prior modeling methodology. First, we follow the non-i.i.d. noise assumption of HSI [22] and utilize the NMoG to characterize the noise structure, which is equivalent to optimizing a robust weighted loss function. Second, we fully exploit the prior knowledge of HSI by considering the nonlocal similarity prior, the spatial–spectral correlation prior, and the nonlocal smoothness prior. Specifically, we apply the nonlocal low-rank tensor model to model the nonlocal similarity and the spatial–spectral correlation prior. Besides, we design an adaptive edge-preserving TV regularization term to characterize the nonlocal smooth prior of HSI. Then, we develop an effective alternating direction method of multipliers (ADMM) algorithm to solve the proposed model. Finally, experimental results on both synthetic and real HSI datasets illustrate that our proposed method can achieve better performance compared with other state-of-the-art methods.

The rest of the article is organized as follows: The notations are presented in Section II. The proposed method is presented in Section III. The optimization algorithm is proposed in Section IV. Experimental results are shown in Section V. Finally, conclusions are drawn in Section VI.

II. Notations and Preliminaries

A. Notations

The tensor is treated as a multiindex array, and its order is defined as the number of its modes or dimensions. A tenor with order N can be represented by decorated letters $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ and its elements is denoted by $\mathcal{X}_{i_1,i_2,\ldots,i_N}$. The tensor can be unfolded into a matrix among mode-n, which denoted as $\mathrm{X}_{(n)}$, where the element in $\mathcal{X}$ located in $(i_1, i_2, \ldots, i_n)$ is mapped to the element in $\mathrm{X}_{(n)}$ located in (i_n, j), where $j = 1 + \sum_{k=1,k\neq n}^{N} (i_k - 1) \prod_{m=1,m\neq n}^{k-1} I_m$. The mode-$n$ multiplication of tensor $\mathcal{X}$ and matrix U is denoted by $\mathcal{X} \times_n \mathrm{U}$, where the element $(\mathcal{X} \times_n \mathrm{U})_{i_1,\ldots,i_{n-1},j,i_{n+1},\ldots,i_N} = \sum_{i_n} x_{i_1,\ldots,i_N} \times u_{j,i_n}$. Moreover, we denote matrices with nonbold upper-case letters X, the vectors with bold lower-case letters $\boldsymbol{x}$, and the scalars nonbold letters x, respectively.

B. Non-i.i.d. Noise Modeling

The non-i.i.d. noise modeling methodology is first proposed in [22], which claims that the complex HSI noise (i.e., Gaussian noise, stripe noise, deadline noise, and impulse noise) has the following characteristics: 1) noise distribution in each band is usually consistent; 2) noise intensity and noise type in different bands vary greatly; and 3) noise in different bands also shares some common characteristic since all the bands are captured by the same imaging system.

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Based on these characteristics, the noise located in each band was assumed to be generated from an independent MoG distribution due to its universal approximation capability for any continuous densities [65], which has been extensively studied in [61] and [66]. To encode the variation of distribution among bands, the parameters of each MoG model were assumed to be different. Therefore, the distribution of noise located in ith row, jth column, and bth band was modeled as

$$p(\mathcal{E}_{ijb}) = \sum_{k=1}^{K} \pi_{bk} \mathrm{N}(\mathcal{E}_{ijb} | \mu_{bk}, \gamma_{bk}^{-1}) \tag{1}$$

where $\mathcal{E}_{ijb}$ denotes the element located in ith row, jth column, and bth band of $\mathcal{E}$, π_{bk} is the mixing proportion with $\pi_{bk} \geq 0$ and $\sum_{k=1}^{K} \pi_{bk} = 1$, K is the Gaussian component number, and γ_{bk} is the precision of kth Gaussian component in the bth band. To encode the common characteristics of noise between different bands, the parameter $\boldsymbol{\gamma}_b$ of each MoG model was assumed to generate from a two-level prior distribution

$$\begin{aligned} \mu_{dk}, \gamma_{dk} &\sim \mathrm{N}(\mu_{dk} | m_0, (\beta_0 \gamma_{dk})^{-1}) \mathrm{Gam}(\gamma_{dk} | c_0, d) \\ d &\sim \mathrm{Gam}(d | \eta_0, \lambda_0) \end{aligned} \tag{2}$$

where $\mathrm{Gam}(\cdot)$ represents the Gamma distribution, and c_0, η_0, and λ_0 are preset parameters of Gamma distribution.

This NMoG noise model reasonably characterizes the characteristic of HSI noise, where the posterior distributions of all involved parameters are efficiently solved with a closed-form solution by variational inference.

III. Proposed Method

In this section, we first present the HSI denoising problem and the non-i.i.d. noise modeling methodology. Then, we introduce our proposed adaptive TV regularization term. Finally, we introduce our proposed HSI denoising model.

A. Problem Formulation

HSI can be represented as a 3-D tensor. Due to the effect of thermal electronics, dark current, and stochastic error in the imaging process, HSIs are always corrupted by complex noise. For simplicity, we assume the noise is additive, and thus the noise degradation model of an HSI can be expressed as

$$\mathcal{Y} = \mathcal{X} + \mathcal{E} \tag{3}$$

where $\mathcal{Y} \in \mathbb{R}^{M \times N \times B}$, $\mathcal{X} \in \mathbb{R}^{M \times N \times B}$, and $\mathcal{E} \in \mathbb{R}^{M \times N \times B}$ denote the observed noisy HSI, the clean his, and the noise, respectively. M and N are the height and width of the spatial image, respectively, and B is the number of spectral bands. The HSI denoising problem is to recover $\mathcal{X}$ from $\mathcal{Y}$.

B. Weighted Fidelity Function

We inherit and simplify the NMoG noise model by assuming the mean of noise is 0, so the distribution of noise can be modeled as

$$\begin{aligned} p(\mathcal{E}_{ijb}) &= \sum_{k=1}^{K} \pi_{bk} \mathrm{N}(\mathcal{E}_{ijb} | 0, \gamma_{bk}^{-1}) \\ \gamma_{bk} &\sim \mathrm{Gam}(\gamma_{bk} | c_0, d_{bk}) \\ d_{bk} &\sim \mathrm{Gam}(d_{bk} | \eta_0, \lambda_0). \end{aligned} \tag{4, 5}$$

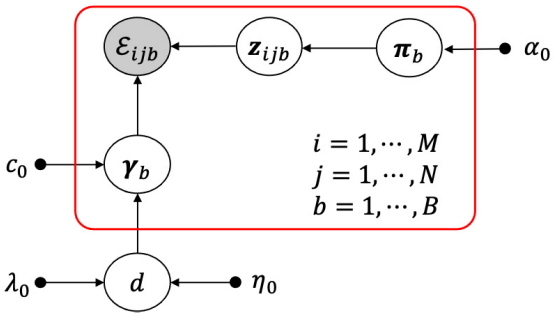


Fig. 1. Graphical model of noise modeling. $\mathcal{E}_{ijb}$ denotes the noise located in ith row jth column and bth band of noise tensor $\mathcal{E}$. $\boldsymbol{\gamma}_b$ is the k-dimensional vector representing the precision of all MoG components in the bth band, with hyperparameters d, c_0, where d is with hyperparameters η_0, λ_0. z_{ijb} is the hidden variable vector generated from multinomial distribution with parameter $\boldsymbol{\pi}_b$, with hyperparameter α_0.

In order to apply the variational expectation maximization (VEM) algorithm to solve the model, we need to introduce a latent variable $\mathcal{Z}$, and then (4) can be equivalently rewritten as

$$\begin{aligned} \mathcal{E}_{ijb} &\sim \prod_{k=1}^{K} \mathrm{N}(\mathcal{E}_{ijb} | 0, \gamma_{bk}^{-1})^{z_{ijbk}} \\ z_{ijb} &\sim \mathrm{Mul}(z_{ijb} | \boldsymbol{\pi}_b) \\ \boldsymbol{\pi}_b &\sim \mathrm{Dir}(\boldsymbol{\pi}_b | \alpha_0) \end{aligned} \tag{6}$$

where $\mathrm{Mul}(\cdot)$ and $\mathrm{Dir}(\cdot)$ represent the multinomial distribution and Dirichlet distribution, respectively. Therefore, the non-i.i.d. noise structure of the HSI is characterized by (5) and (6). The graphical model of this noise modeling methodology is illustrated in Fig. 1. All involved parameters can be inferred from data and this will be discussed in detail in Section IV.

Based on the graphical model shown in Fig. 1, the complete log-likelihood can be expressed as (7), shown at the bottom of the next page, where $\mathcal{Z} = \{z_{ijb}\}$, $\Pi = \{\pi_{bk}\}$, $\Gamma = \{\gamma_{bk}\}$, and $\boldsymbol{d} = \{d_{bk}\}$. To apply the VAE algorithm to maximize the log-likelihood of the observed data, we need to calculate the complete log-likelihood first as shown in (7). Additionally, we need to approximate the posterior distribution $p(\mathcal{Z}, \Pi, \Gamma, d | \mathcal{X})$ by using the variational method. All these details will be discussed in Section IV. In the M step of the VEM algorithm, we need to maximize the expectation of the complete log-likelihood with respect to $\mathcal{X}$, the loss function of this optimization problem can be reformulated as follows:

$$\begin{aligned} \mathrm{Loss}(\mathcal{Y}, \mathcal{X}) &= \| \mathcal{W} \odot (\mathcal{Y} - \mathcal{X}) \|_F^2 \\ \mathcal{W}_{ijb} &= \sqrt{\sum_{k=1}^{K} \frac{1}{2} \langle \mathcal{Z}_{ijbk} \rangle \langle \gamma_{bk} \rangle} \end{aligned} \tag{8}$$

where $\odot$ represents the element-wise product.

C. Adaptive Total Variation Regularization

The TV model is an effective and efficient way to encode the local smoothness prior, which is widely used in image denoising. Compare to a 2-D image, HSI shows local smoothness not only in the spatial domain but also in the spectral

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domain [29], [31], [47]. The spatial–spectral TV (SSTV) of HSI can be described as

$$\|\mathcal{X}\|_{\mathrm{SSTV}} = \|\boldsymbol{D}\mathcal{X}\|_1 = \|\boldsymbol{D}_h\mathcal{X}\|_1 + \|\boldsymbol{D}_v\mathcal{X}\|_1 + \|\boldsymbol{D}_s\mathcal{X}\|_1 \quad (9)$$

where $\boldsymbol{D} = [\boldsymbol{D}_h, \boldsymbol{D}_v, \boldsymbol{D}_s]$ denotes the 3-D first-order forward finite-difference operator. $\boldsymbol{D}_h$, $\boldsymbol{D}_v$, and $\boldsymbol{D}_s$ denote the first-order finite-difference operators along the spatial horizontal, spatial vertical, and spectral mode, respectively, which are defined as follows:

$$\begin{aligned}\boldsymbol{D}_h\mathcal{X} &= \mathcal{X}_{i,j+1,b} - \mathcal{X}_{i,j,b},\\ \boldsymbol{D}_v\mathcal{X} &= \mathcal{X}_{i+1,j,b} - \mathcal{X}_{i,j,b},\\ \boldsymbol{D}_s\mathcal{X} &= \mathcal{X}_{i,j,b+1} - \mathcal{X}_{i,j,b}.\end{aligned} \quad (10)$$

From the definition of SSTV, we can see that the distributions of different values in all directions as treated the same, which deviates greatly from the real scene. In fact, the noise intensity and pixel scale values of each band are different, and thus the distributions of difference values along three modes are different [31], [47]. Such SSTV will result in oversmoothed in some bands and weak noise suppressed in some other bands. Additionally, in the spatial domain, the edge of the texture of the image should be protected from being smoothed.

To alleviate these issues, some adaptive SSTV (ASSTV) models based on weighted strategy have been proposed [29], [47]. We inherited their idea to propose a new ASSTV regularization, which is defined as follows:

$$\begin{aligned}\|\mathcal{X}\|_{\mathrm{ASSTV}} &= \|\mathcal{S} \odot (\boldsymbol{D}\mathcal{X})\|_1\\ &= \|\mathcal{S}_h \odot (\boldsymbol{D}_h\mathcal{X})\|_1 + \|\mathcal{S}_v \odot (\boldsymbol{D}_v\mathcal{X})\|_1\\ &\quad + \|\mathcal{S}_s \odot (\boldsymbol{D}_s\mathcal{X})\|_1\end{aligned} \quad (11)$$

where $\mathcal{S} = [\mathcal{S}_h, \mathcal{S}_v, \mathcal{S}_s] \in \mathbb{R}^{M\times N\times 3B}$ is the weight tensor that is used to adjust the smoothing strength in different pixel locations, and $\mathcal{S}_h, \mathcal{S}_v, \mathcal{S}_s$ denote the weights to difference values along spatial horizontal, spatial vertical, and spectral mode, respectively.

How to determine weight tensor $\mathcal{S}$ is the key to our ASSTV model. Considering that if clean HSI $\mathcal{X}$ is known, then $\boldsymbol{D}\mathcal{X}$ accurately reflects the edge information in the image space and the scale information of the pixel values for different spectral bands. We can directly use normalized $\boldsymbol{D}\mathcal{X}$ to determine the weight tensor $\mathcal{S}$. However, the clean HSI data is unknown. Fortunately, we can obtain the denoised data during the denoising process. Denote the denoised HSI data as $\hat{\mathcal{X}}$, in order to avoid the interference of residual noise, we carry out further processing on $\hat{\mathcal{X}}$ as follows:

$$\mathcal{X}_0 = \mathrm{LRA}_1\left(\hat{\mathcal{X}}\right) \quad (12)$$

where $\mathrm{LRA}_1(\cdot)$ means the rank-1 low-rank approximation of $\hat{\mathcal{X}}$. Thus, $\mathcal{X}_0$ contains only the most important component of $\mathcal{X}$, and the influence of noise is minimized. Then, the weight tensor $\mathcal{S}$ can be computed as

$$\begin{aligned}\mathcal{S}_h &= \frac{\max(|\boldsymbol{D}_h\mathcal{X}_0|)}{|\boldsymbol{D}_h\mathcal{X}_0| + \delta\max(|\boldsymbol{D}_h\mathcal{X}_0|)}\\ \mathcal{S}_v &= \frac{\max(|\boldsymbol{D}_v\mathcal{X}_0|)}{|\boldsymbol{D}_v\mathcal{X}_0| + \delta\max(|\boldsymbol{D}_v\mathcal{X}_0|)}\\ \mathcal{S}_s &= \frac{\max(|\boldsymbol{D}_s\mathcal{X}_0|)}{|\boldsymbol{D}_s\mathcal{X}_0| + \delta\max(|\boldsymbol{D}_s\mathcal{X}_0|)}\end{aligned} \quad (13)$$

where δ is the threshold used to avoid undesirable large values in the weight tensor.

D. WNLRATV Model

As mentioned previously, the nonlocal spatial similarity and the spatial and spectral correlation are important prior knowledge for HSI restoration. The existing denoising methods focus more on utilizing these priors but underestimate the noise complexity in a real scene. For example, they usually assume that the noise is independent and identically distributed (i.i.d.) Gaussian noise, or Gaussian noise mixed with sparse noise. Such noise assumptions are inconsistent with the real situation and thus result in less robustness in removing complex noise.

In this section, we propose a new denoising method by combining the advantages of noise modeling and prior knowledge encoding. First, we fully consider the complexity of noise and obtain a robust loss metric through the noise modeling method. Then, we encode the most important prior knowledge, that is, the spatial smoothness, the nonlocal spatial similarity, and the spatial and spectral correlation, as regularization of clean HSI. Finally, we integrate the robust loss metric and these regularizations into the following model:

$$\min_{\mathcal{X}} \|\mathcal{W} \odot (\mathcal{Y} - \mathcal{X})\|_F^2 + \lambda_1\|\mathcal{S} \odot (\boldsymbol{D}\mathcal{X})\|_1 + \lambda_2\|\mathcal{X}\|_{\mathrm{NSS}} \quad (14)$$

where $\|\mathcal{X}\|_{\mathrm{NSS}}$ is the regularization to encode the nonlocal spatial similarity and the spatial–spectral correlation. There have been many excellent works on this area, such as low-rank tensor decomposition [42], [43], [47], or patch-based low-rank methods [37], [48], [67]. All these methods perform well in HSI denoising tasks. Among them, the patch-based low-rank approximation method can better model the local characteristics of HSI, and thus we use a patch-based low-rank matrix approximation model to design the regular term $\|\mathcal{X}\|_{\mathrm{NSS}}$. Since some HSIs have relatively high spectral dimensionality, the calculation of full-band group matching will be time-consuming. In order to calculate efficiently, we follow the idea in [37] to encode $\|\mathcal{X}\|_{\mathrm{NSS}}$. Specifically, since HSI lies in a low-dimensional spectral subspace due to the global spectral correlation characteristic, we first learn the basics of the intrinsic spectral subspace and then apply the nonlocal

$$\begin{aligned}p(\mathcal{Y}, \mathcal{Z}, \Pi, \Gamma, \boldsymbol{d}|\mathcal{X}) &= \prod p(\mathcal{Y}|\mathcal{X}, \mathcal{Z}, \Gamma)\, p(\mathcal{Z}|\Pi)p(\Pi)p(\Gamma|\mathrm{d})\, p(\mathrm{d})\\ &= \prod_{i,j,b,k} \mathrm{N}\left(\mathcal{Y}_{ijb}|\mathcal{X}_{ijb}, \gamma_{bk}^{-1}\right)^{\mathcal{Z}_{ijbk}} \prod_{i,j,b} \mathrm{Mul}\left(\boldsymbol{z}_{ijb}|\boldsymbol{\pi}_b\right)\\ &\quad \prod_b \mathrm{Dir}\left(\boldsymbol{\pi}_b|\alpha_0\right) \prod_{b,k} \mathrm{Gam}\left(\gamma_{bk}|c_0, d\right)\mathrm{Gam}(d_{bk}|\eta_0, \lambda_0)\end{aligned} \quad (7)$$

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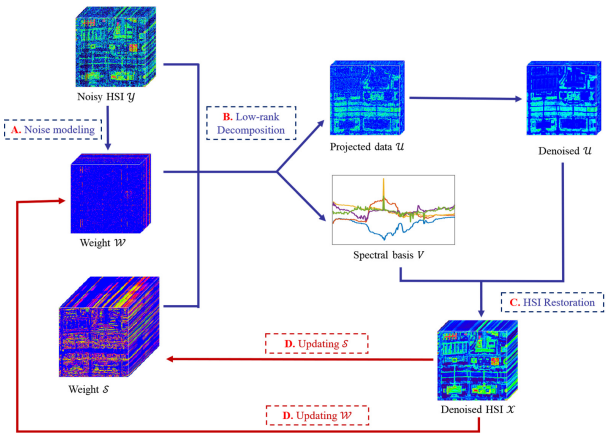


Fig. 2. Flowchart of the proposed method. It mainly includes four stages. A: Learning the weight tensor $\mathcal{W}$ involved in weighted fidelity function by noise modeling. B: Solving the weighted nonlocal low-rank model with adaptive total regularization to update the low-rank components $\mathcal{U}$ and V. C: Restoring HSI data $\mathcal{X}$. D: Updating the weight tensors $\mathcal{W}$ and $\mathcal{S}$ with restored data $\mathcal{X}$.

low-rank tensor model to the projected data. Based on this modeling of $\|\mathcal{X}\|_{\text{NSS}}$, we obtain another realization of the method, the corresponding optimization problem is expressed as follows:

$$\begin{aligned}\min_{\mathcal{X},\mathcal{U},\text{V}} &\|\mathcal{W}\odot(\mathcal{Y}-\mathcal{X})\|_F^2+\lambda_1\|\mathcal{S}\odot(\boldsymbol{D}\mathcal{X})\|_1\\ &+\lambda_2\sum_{i=1}^{P}\|\boldsymbol{P}_i\mathcal{U}\|_{w,*}\\ \text{s.t.}\quad &\mathcal{X}=\mathcal{U}\times_3\text{V}\end{aligned}\tag{15}$$

where V and $\mathcal{U}$ denote the basis of spectral subspace and the projection of $\mathcal{X}$ onto the spectral subspace.

So far, we have introduced the optimization problem corresponding to the model of the proposed method, and the overview of the proposed paradigm is shown in Fig. 2. Then, we will introduce the solution process in detail in the following section.

IV. Optimization Algorithm

A. Adaptive Learning of Loss Function

In this section, we first introduce the deduction of the loss function and weights $\mathcal{W}$ involved in it as shown in (8).

The full likehood $p(\mathcal{Y},\mathcal{Z},\Pi,\Gamma,\boldsymbol{d}|\mathcal{X})$ is expressed in (7). We denote the parameters of noise distribution as $\Theta=(\mathcal{Z},\Pi,\Gamma,\boldsymbol{d})$. The VEM algorithm is used to maximize the likelihood function $p(\mathcal{Y}|\mathcal{X})$ with respect to $\mathcal{X}$ and is composed of two steps:

1) V Step: We use current estimated data $\mathcal{X}^{\text{old}}$ to evaluate the posterior distribution of Θ, that is, $q(\Theta|\mathcal{Y},\mathcal{X}^{\text{old}})$. Once $\mathcal{Y}$ and $\mathcal{X}^{\text{old}}$ are known, the estimated noise data can be obtained as follows:

$$\mathcal{E}=\mathcal{Y}-\mathcal{X}^{\text{old}}.\tag{16}$$

We apply the variational inference method to approximate $p(\Theta|\mathcal{Y},\mathcal{X}^{\text{old}})$, assuming the approximate posterior of each parameter is independent and thus has the factorized form as follows:

$$q(\mathcal{Z},\Pi,\Gamma,\boldsymbol{d})=\prod_{ijb}q(\mathcal{Z}_{ijb})\prod_{b}q(\boldsymbol{\pi_b})\prod_{bk}q(\boldsymbol{\gamma}_{bk})q(\boldsymbol{d}_{bk}).$$

The approximate posteriors have closed-form solutions, and the details are shown in the Appendix. Here, we give the updating equations directly. The posterior distribution of precision $\boldsymbol{\gamma}_b$ for noise in each band is updated by the following equation:

$$q^*(\boldsymbol{\gamma}_b)=\prod_k\text{Gam}(\gamma_{bk}|c_{bk},d_{bk})\tag{17}$$

where the parameters in the above equation can be calculated from data in the following way:

$$c_{bk}=c_0+\frac{1}{2}\sum_{ij}\langle\mathcal{Z}_{ijbk}\rangle\tag{18}$$

$$d_{bk}=\langle d_{bk}\rangle+\frac{1}{2}\left\{\sum_{ij}\mathcal{E}_{ijb}^2\langle\mathcal{Z}_{ijbk}\rangle\right\}.\tag{19}$$

The update equation of the latent variable $\mathcal{Z}$ is

$$q^*(\mathcal{Z}_{ijb})=\prod_k\varrho_{ijbk}^{\mathcal{Z}_{ijbk}}\tag{20}$$

where the involved parameters are calculated by

$$\begin{aligned}\varrho_{ijbk}&=\rho_{ijbk}/\sum_k\rho_{ijbk}\\ \ln\rho_{ijbk}&=\langle\ln\pi_{bk}\rangle-\ln\sqrt{2\pi}+\langle\ln\gamma_{bk}\rangle/2\\ &\quad-\mathcal{E}_{ijb}^2\langle\gamma_{bk}\rangle/2.\end{aligned}\tag{21}$$

Similarly, the update equation for the mixing proportion $\boldsymbol{\pi}_b$ over the bth band can be written as

$$q^*(\boldsymbol{\pi}_b)=\prod_k\pi_{bk}^{\alpha_{bk}-1}\tag{22}$$

where $\alpha_{bk}=\alpha_0+\sum_{ij}\langle\mathcal{Z}_{ijbk}\rangle$.

The update equation on the hyperparameter $\boldsymbol{d}$ is

$$q^*(d_{bk})=\text{Gam}(d_{bk}|\eta,\lambda)\tag{23}$$

where $\eta=\eta_0+c_0KB$ and $\lambda=\lambda_0+\langle\gamma_{bk}\rangle$.

2) E Step: We use the posterior distribution to find the expectation of complete-data log likelihood [68]

$$\begin{aligned}Q(\mathcal{X},\mathcal{X}^{\text{old}})&=\int q(\Theta|\mathcal{Y},\mathcal{X}^{\text{old}})\ln p(\Theta,\mathcal{X}|\mathcal{Y})d\Theta\\ &=-\sum_{i,j,b,k}\frac{1}{2}\langle z_{ijbk}\rangle\langle\gamma_{bk}\rangle\left(\mathcal{Y}_{ijb}-\mathcal{X}_{ijb}\right)^2+\text{const}\end{aligned}\tag{24}$$

where the constant represents the polynomial, which is not related to $\mathcal{X}$.

3) M Step: We update $\mathcal{X}$ by maximizing $Q(\mathcal{X},\mathcal{X}^{\text{old}})$, which can be rewritten equivalently as follows:

$$\begin{aligned}Q(\mathcal{X},\mathcal{X}^{\text{old}})&=-\|\mathcal{W}\odot(\mathcal{Y}-\mathcal{X})\|_F^2\\ \mathcal{W}_{ijb}&=\sqrt{\sum_{k=1}^{K}\frac{1}{2}\langle\mathcal{Z}_{ijbk}\rangle\langle\gamma_{bk}\rangle}.\end{aligned}\tag{25}$$

So far, the derivation of the loss function and the calculation of involved parameters have been completed.

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B. Optimization of WNLRATV

For the WNLRATV, let the constraint strictly true, and then the original problem (15) can be rewritten as the following optimization problem by introducing some auxiliary variables:

$$\begin{aligned}\min_{\mathcal{U},\mathrm{V},\mathcal{A}} &\|\mathcal{W}\odot(\mathcal{Y}-\mathcal{U}\times_3\mathrm{V})\|_F^2+\lambda_1\|\mathcal{S}\odot\mathcal{A}\|_1\\ &+\lambda_2\sum_{i=1}^{P}\|\boldsymbol{P}_i\mathcal{U}\|_{w,*}\\ \text{s.t.}\quad &\mathcal{C}=\mathcal{U}\times_3\mathrm{V}\\ &\mathcal{A}=\boldsymbol{D}\mathcal{C}.\end{aligned}\tag{26}$$

Based on the ADMM methodology, the above optimization problem can be transformed into minimizing the following augmented Lagrangian function:

$$\begin{aligned}L(\mathcal{U},\mathrm{V},\mathcal{C},\mathcal{A},\mathcal{M}_1,\mathcal{M}_2)=&\|\mathcal{W}\odot(\mathcal{Y}-\mathcal{U}\times_3\mathrm{V})\|_F^2\\ &+\lambda_1\|\mathcal{S}\odot\mathcal{A}\|_1+\lambda_2\sum_{i=1}^{P}\|\boldsymbol{P}_i\mathcal{U}\|_{w,*}\\ &+\frac{\rho_1}{2}\|\mathcal{U}\times_3\mathrm{V}+\frac{\mathcal{M}_1}{\rho_1}-\mathcal{C}\|_F^2\\ &+\frac{\rho_2}{2}\|\boldsymbol{D}\mathcal{C}+\frac{\mathcal{M}_2}{\rho_2}-\mathcal{A}\|_F^2\end{aligned}\tag{27}$$

where $\mathcal{M}_1$ and $\mathcal{M}_2$ are the Lagrange multipliers, and ρ_1 and ρ_2 are the penalty parameters.

In the ADMM framework, we can alternatively optimize the augmented Lagrangian function over one variable while keeping all others fixed. We then discuss the solution to the subproblems with respect to each involved variable.

1) Update $\mathcal{U}$ and V*:* The solution is divided into two steps. First, we need to learn the subspace basis V and projected data $\mathcal{U}$ by solving the following weighted l_2-norm low-rank matrix factorization problem:

$$\left\{\mathcal{U}^{(t+\frac{1}{2})},\mathrm{V}^{(t+1)}\right\}=\arg\min_{\mathcal{U},\mathrm{V}}\|\tilde{\mathrm{W}}_{(3)}\odot\left(\tilde{\mathrm{Y}}_{(3)}-\mathrm{U}_{(3)}\mathrm{V}\right)\|_F^2\tag{28}$$

where

$$\begin{aligned}\tilde{\mathcal{W}}&=\sqrt{\mathcal{W}^2+\frac{\rho_1}{2}}\\ \tilde{\mathcal{Y}}&=\frac{\mathcal{W}^2\odot\mathcal{Y}+\frac{\rho_1}{2}\left(\mathcal{C}-\frac{\mathcal{M}_1}{\rho_1}\right)}{\mathcal{W}^2+\frac{\rho_1}{2}}.\end{aligned}\tag{29}$$

The alternated least-square algorithm [69] can be directly applied to solve the problem. Then, the projected data $\mathcal{U}$ can be denoised by solving the following subproblem:

$$\min_{\mathcal{U}}\|\mathcal{U}^{(t+\frac{1}{2})}-\mathcal{U}\|_F^2+\lambda_2\sum_{i=1}^{P}\|\boldsymbol{P}_i\mathcal{U}\|_{w,*}.\tag{30}$$

The solution of $\mathcal{U}$ can be obtained as follows:

$$\begin{aligned}\mathcal{U}^{(t+1)}&=\boldsymbol{P}^{-1}\left(\{\mathrm{L}_i\}_{i=1}^{P}\right)\\ \mathrm{L}_i&=\mathrm{U}\hat{\Sigma}\hat{\mathrm{V}}^T\end{aligned}\tag{31}$$

where U and $\hat{\mathrm{V}}$ are the right and left singular matrices of $\widetilde{\mathrm{Y}}=\boldsymbol{P}_i\mathcal{U}^{(t+\frac{1}{2})}$, respectively. $\hat{\Sigma}$ is the diagonal matrix, and the diagonal elements are calculated as follows:

$$\sigma_i=\begin{cases}0\quad, & \text{if}\quad c_2<0\\ \dfrac{c_1+\sqrt{c_2}}{2}, & \text{if}\quad c_2\geqslant 0.\end{cases}\tag{32}$$

The parameters involved in this formula can be calculated as follows:

$$\begin{aligned}c_1&=\sigma_i\left(\widetilde{\mathrm{Y}}\right)-\epsilon\\ c_2&=\left(\sigma_i\left(\widetilde{\mathrm{Y}}\right)+\epsilon\right)^2-4\sqrt{2}.\end{aligned}\tag{33}$$

2) Update $\mathcal{C}$: Extracting all terms containing $\mathcal{C}$ from (26), the subproblem for updating $\mathcal{C}$ can be expressed as follows:

$$\min_{\mathcal{C}}\|\mathcal{X}+\frac{1}{\rho_1}\mathrm{M}_1-\mathcal{C}\|_F^2+\frac{\rho_2}{\rho_1}\|\mathcal{A}-\frac{1}{\rho_2}\mathrm{M}_2-\boldsymbol{D}\mathcal{C}\|_F^2.\tag{34}$$

This problem can be treated as solving the linear system

$$\left(\mathrm{I}+\frac{\rho_2}{\rho_1}\boldsymbol{D}^*\boldsymbol{D}\right)\mathcal{C}=\mathcal{X}+\frac{1}{\rho_1}\mathrm{M}_1+\frac{\rho_2}{\rho_1}\boldsymbol{D}^*\left(\mathcal{A}-\frac{1}{\rho_2}\mathrm{M}_2\right)\tag{35}$$

where $\boldsymbol{D}^*$ denotes the adjoint operator of $\boldsymbol{D}$. We can apply a fast Fourier transform to solve this problem, and the closed-form solution to $\mathcal{C}$ is deduced as follows:

$$\begin{cases}\mathcal{H}=\mathcal{U}\times_3\mathrm{V}+\dfrac{1}{\rho_1}\mathcal{M}_1+\dfrac{\rho_2}{\rho_1}\boldsymbol{D}^*\left(\mathcal{A}-\dfrac{1}{\rho_2}\mathrm{M}_2\right)\\ \mathcal{T}=|\mathrm{fftn}(\boldsymbol{D}_h)|^2+|\mathrm{fftn}(\boldsymbol{D}_v)|^2+|\mathrm{fftn}(\boldsymbol{D}_s)|^2\\ \mathcal{C}^{t+1}=\mathrm{ifftn}\left(\dfrac{\mathrm{fftn}(\mathcal{H})}{\mathbf{1}+\frac{\rho_2}{\rho_1}\mathcal{T}}\right)\end{cases}\tag{36}$$

where $|\cdot|^2$ denotes the element-wise square, fftn and ifftn represent the fast 3-D Fourier transform and it's inverse transform, respectively, and $\mathbf{1}$ represents the tensor with all elements 1.

3) Update $\mathcal{A}$: Extracting the terms containing $\mathcal{A}$ from (15), the subproblem with respect to $\mathcal{A}$ can be expressed as

$$\min_{\mathcal{A}}\frac{1}{2}\|\boldsymbol{D}\mathcal{C}+\frac{1}{\rho_2}\mathrm{M}_2-\mathcal{A}\|_F^2+\frac{\lambda_1}{\rho_2}\|\mathcal{S}\odot\mathcal{A}\|_1.\tag{37}$$

The closed-form solution of $\mathcal{A}$ can be obtained by using the known *soft-thresholding* operator [70] as follows:

$$\mathcal{A}_{ijb}^{(t+1)}=\boldsymbol{R}_{\frac{\lambda_1\mathcal{S}_{ijb}}{\rho_2}}\left(\boldsymbol{D}\mathcal{C}+\frac{1}{\rho_2}\mathrm{M}_2\right)\tag{38}$$

where $\boldsymbol{R}_\lambda(\cdot)$ represents the *soft-thresholding* operator with parameter λ.

4) Update $\mathcal{X}$: The restored HSI data $\mathcal{X}$ can be updated as follows:

$$\mathcal{X}^{(t+1)}=\mathcal{U}\times_3\mathrm{V}.\tag{39}$$

The multiplier can be updated by the following equations:

$$\begin{cases}\mathcal{M}_1^{(t+1)}=\mathcal{M}_1^{(t+1)}+\rho_1(\mathcal{U}\times_3\mathrm{V}-\mathcal{C})\\ \mathcal{M}_2^{(t+1)}=\mathcal{M}_2^{(t+1)}+\rho_2(\boldsymbol{D}\mathcal{C}-\mathcal{A}).\end{cases}\tag{40}$$

The proposed algorithm for the WNLRATV can then be summarized in Algorithm 1.

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TABLE I
PERFORMANCE COMPARISON OF ALL COMPETING METHODS ON INDIAN PINES DATA

	Noisy HSI	BM4D	LRTV	NMoG	LRTDTV	NGmeet	NMoG-Tucker	QRNN	SQAD	WNLRATV
i.i.d. Gaussian Noise										
MPSNR	20.172	38.592	38.389	34.446	40.854	**45.242**	34.547	34.644	31.014	43.643
MSSIM	0.373	0.977	0.989	0.887	0.989	**0.997**	0.890	0.953	0.923	0.994
ERGAS	14.362	1.784	1.846	2.909	1.448	**0.840**	2.878	3.044	4.565	1.060
SAM	0.194	0.021	0.017	0.035	0.016	**0.008**	0.035	0.032	0.044	0.013
Time	0.000	126.229	107.555	32.136	165.257	64.904	417.622	0.274	0.705	267.492
Non i.i.d. Gaussian Noise										
MPSNR	18.715	35.969	36.779	33.982	39.667	38.780	33.944	32.672	30.351	**39.927**
MSSIM	0.337	0.944	0.980	0.873	0.982	**0.989**	0.878	0.923	0.887	**0.989**
ERGAS	18.722	2.581	2.918	3.291	**1.574**	1.844	2.963	3.754	4.824	1.662
SAM	0.263	0.034	0.038	0.043	**0.018**	**0.018**	0.036	0.042	0.051	0.020
Time	0.000	131.881	119.837	82.423	220.514	93.365	429.024	0.274	0.695	354.895
Gaussian + Stripe Noise										
MPSNR	18.447	33.395	36.192	33.547	38.346	37.931	33.415	32.377	30.163	**39.081**
MSSIM	0.331	0.901	0.981	0.871	0.983	0.988	0.871	0.922	0.885	**0.989**
ERGAS	18.906	3.515	2.585	3.223	1.836	2.018	3.158	3.861	4.897	**1.725**
SAM	0.265	0.047	0.031	0.040	0.021	0.021	0.039	0.043	0.053	**0.020**
Time	0.000	125.352	102.745	51.909	168.966	66.265	494.380	0.274	0.702	226.124
Gaussian + Deadline noise										
MPSNR	16.832	27.328	34.535	32.585	36.806	28.805	32.707	31.591	25.692	**37.877**
MSSIM	0.303	0.793	0.973	0.871	0.979	0.854	0.870	0.898	0.797	**0.982**
ERGAS	23.847	13.937	5.070	4.013	2.538	7.893	3.631	4.152	8.319	**2.477**
SAM	0.334	0.179	0.067	0.051	**0.029**	0.096	0.045	0.047	0.087	0.032
Time	0.000	120.444	108.235	71.590	212.688	79.940	525.889	0.274	0.694	332.672
Mixture Noise										
MPSNR	15.903	27.403	34.745	31.614	36.568	30.243	31.658	31.162	26.854	**36.731**
MSSIM	0.274	0.818	0.971	0.851	0.978	0.922	0.856	0.896	0.804	**0.981**
ERGAS	25.097	10.125	4.750	4.056	**2.229**	5.437	3.923	4.336	7.000	2.367
SAM	0.346	0.127	0.065	0.050	**0.025**	0.059	0.047	0.049	0.075	0.028
Time	0.000	173.904	117.404	89.837	290.346	83.555	625.729	0.274	0.696	385.533

Algorithm 1 Algorithm for WNLRATV

Input: the observed HSI tensor $\mathcal{Y} \in \mathbb{R}^{M\times N\times B}$, Gaussian component number K, parameters λ_1, λ_2.

Output: The restored HSI $\mathcal{X}$.

Initialization: Parameters $(c_0, d_0, \eta_0, \lambda_0)$ in noise prior. The spectral subspace basis V and projection data $\mathcal{U}$ by SVD of $\mathcal{Y}_{(3)}$. $\mathcal{C} = \mathcal{U} \times_3 \mathrm{V}$, $\mathcal{M}_1$, $\mathcal{M}_2$ are zeros tensor. $t = 1$.

1: **while** not coverged **do**
2: Update $\mathcal{W}$ involved in loss function by Eq. (17)-(25).
3: Update $\mathcal{U}$, V,$\mathcal{C}$, $\mathcal{A}$,$\mathcal{X}$ by Eq. (28)-(39).
4: Update lagrange multipliers $\mathcal{M}_1$, $\mathcal{M}_2$ by Eq. (40).
5: $t \leftarrow t + 1$.
6: **end while**

V. EXPERIMENTS

In this section, we conducted a series of experiments on both synthetic and real HSI data to evaluate the performance of the proposed two WNLRATV models. For comparison, we consider the following state-of-the-art HSI denoising methods including BM4D [24] representing block-matching and 4-D filtering method, LRTV method [35] considering the spectral correlation and nonlocal smoothness prior, LRTDTV method [47] considering the spectral–spatial correlation and nonlocal smoothness prior, NGmeet [37] considering the nonlocal similarity and spectral correlation prior, and NMoG-Tucker method [71] considering the complex noise structure and spectral–spatial correlation prior. Meanwhile, the NMoG method [22] is also compared, which represents the state-of-the-art noise modeling method for HSI denoising. Besides, two deep-learning methods QRNN [59] and SQAD [57] are conducted to provide a baseline comparison with deep HSI prior. In all the following experiments, the parameters of compared methods were manually adjusted according to their default strategies.

A. Simulation Experiments

We conduct the simulated experiment on two HSI data: the simulated ground truth of Indian Pines[1] with size of $145 \times 145 \times 224$ and Washington DC Mall[2] with size of $1208 \times 307 \times 191$. The Washington DC Mall was resized to $200 \times 200 \times 191$ by cropping the main part of the data. In order to simulate the noisy HSI data, we added several types of noise to the original data as follows.

1) *I.i.d. Gaussian noise:* Entries in all bands were corrupted by zero-mean i.i.d. Gaussian noise $\mathrm{N}(0, \sigma^2)$ with the variance $\sigma = 0.1$.

2) *Non-i.i.d. Gaussian noise:* Entries in different bands were corrupted by zero-mean Gaussian noise with different intensities. The signal-to-noise ratio (SNR) value of each band is random with a value in the range of [2, 12] dB.

3) *Gaussian + stripe noise:* All bands were corrupted by non-i.i.d. Gaussian noise as Case (2). Besides, 40% bands were randomly chosen to add stripe noise. The number of stripes in each band is from 20 to 40.

[1] http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes

[2] http://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

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TABLE II
PERFORMANCE COMPARISON OF ALL COMPETING METHODS ON WASHINGTON DC MALL DATA

	Noisy HSI	BM4D	LRTV	NMoG	LRTDTV	NGmeet	NMoG-Tucker	QRNN	SQAD	WNLRATV
i.i.d. Gaussian Noise										
MPSNR	20.172	32.914	35.562	36.192	36.249	**37.475**	36.114	34.888	33.296	37.223
MSSIM	0.448	0.920	0.957	0.967	0.962	**0.973**	0.965	0.957	0.935	0.972
ERGAS	25.934	5.783	4.334	4.082	3.996	**3.431**	4.149	4.760	5.611	3.576
SAM	0.559	0.134	0.109	0.107	0.101	**0.078**	0.106	0.115	0.125	0.090
Time	0.000	156.175	143.842	44.416	299.195	267.452	826.135	0.417	0.740	338.320
Non i.i.d. Gaussian Noise										
MPSNR	21.445	31.234	34.876	37.000	36.068	34.246	37.222	33.960	33.010	**37.326**
MSSIM	0.464	0.817	0.956	0.952	0.946	0.922	**0.973**	0.947	0.931	0.972
ERGAS	24.638	9.169	6.828	5.867	4.491	6.643	**3.699**	5.331	5.898	3.776
SAM	0.609	0.284	0.220	0.205	0.127	0.209	**0.100**	0.143	0.146	0.105
Time	0.000	221.213	133.466	71.629	312.794	381.892	1125.816	0.417	0.722	346.150
Gaussian + Stripe Noise										
MPSNR	20.805	30.147	34.876	36.625	35.619	34.137	36.883	33.712	32.656	**37.158**
MSSIM	0.488	0.853	0.957	0.967	0.958	0.942	**0.971**	0.944	0.927	**0.971**
ERGAS	25.697	9.359	6.780	5.059	4.541	6.423	**3.823**	5.465	6.102	3.827
SAM	0.617	0.276	0.218	0.164	0.120	0.199	**0.103**	0.148	0.152	**0.103**
Time	0.000	148.840	132.861	60.947	286.721	272.084	1220.394	0.416	0.730	331.903
Gaussian + Deadline noise										
MPSNR	20.934	30.147	32.234	35.683	35.483	32.960	36.065	33.760	32.002	**36.411**
MSSIM	0.491	0.864	0.917	0.964	0.958	0.935	0.969	0.945	0.921	**0.970**
ERGAS	25.750	9.312	7.764	5.935	5.168	7.217	4.326	5.489	6.724	**4.302**
SAM	0.635	0.244	0.211	0.157	0.122	0.215	0.111	0.148	0.161	**0.110**
Time	0.000	217.732	164.488	63.207	295.818	207.264	1165.649	0.417	0.726	249.670
Mixture Noise										
MPSNR	19.752	28.835	32.298	34.967	34.572	32.888	35.351	33.200	31.684	**35.572**
MSSIM	0.444	0.826	0.919	0.962	0.949	0.932	**0.965**	0.939	0.914	0.964
ERGAS	28.261	10.907	7.750	5.647	5.810	7.141	**4.570**	5.819	6.919	4.869
SAM	0.655	0.289	0.210	0.165	0.175	0.201	**0.110**	0.151	0.169	0.125
Time	0.000	225.250	162.916	60.770	285.274	205.922	1556.350	0.427	0.723	253.825

4) *Gaussian + deadline noise:* All bands were contaminated by non-i.i.d. Gaussian noise as Case (2). 50% bands on the Indian Pines dataset and 40% bands on the Washington DC Mall dataset were chosen randomly to add deadline noise, respectively. The number of deadlines is 5–25.

5) *Mixture noise:* Each band was randomly corrupted by at least one kind of noise mentioned in Cases (2)–(5).

We performed a Monte Carlo simulation study in the simulation experiment, where 20 realizations were conducted for each noise case. Four criteria were utilized to measure the performance as follows.

1) *MPSNR [72]:* Mean of the peak signal-to-noise ratio (PSNR) over all bands between clean HSI and recovered HSI.
2) *MSSIM [73]:* Mean of structural similarity (SSIM) between clean HSI and recovered HSI over all bands.
3) *ERGAS:* The Erreur Relative Globale Adimensionnellede Synthese.
4) *SAM:* The spectral angle mapper.
5) *Time:* Time required to complete an experiment.

1) Quantitative Comparison: The average measurements of 20 realizations on the Indian Pines dataset were shown in Table I. As can be seen, the proposed method performs more robustly than other compared methods in almost all cases. Although the time cost of our method is relatively large, it is still within an acceptable range, and considering its excellent performance, it should be rational to say that the proposed method is efficient. Moreover, the nonlocal smoothness prior and the nonlocal similarity prior are significant on this dataset, thus the methods that encoded these two priors in the model have better performance, such as LRTDTV and WNLRATV. Note that the deep-learning denoising methods do not perform as well as traditional machine-learning methods, mainly because the dataset is not included in the training dataset, and thus it cannot be recovered accurately.

Table II presents the denoised quantitative results on the Washington DC Mall dataset. In the complex noise case, the proposed method performs either best or slightly worse than the NMoG-Tucker method. However, the time cost of the NMoG-Tucker method is much higher than our method, so it can be said that our method still has significant advantages over all methods. Additionally, the deep-learning denoising methods perform robustly in all noise cases, but their denoising results are weaker than traditional machine-learning methods due to the limited method generalization.

2) Visual Comparison: As a specific example of the Monte-Carlo experiment, the denoising results obtained in one realization of mixture noise case on the Indian Pines dataset and Washington DC Mall dataset were shown in Figs. 3 and 4, respectively. For better visual comparison, we enlarge one common region of each figure, which is marked by a red box. From Fig. 3, it can be easily observed that there still have visually visible noise in the denoising results of comparison methods except the proposed method. From Fig. 4, it can be seen that our method can better remove the noise and preserve the detailed information. Consequently, our method achieves comparable visual denoising performance compared with other methods.

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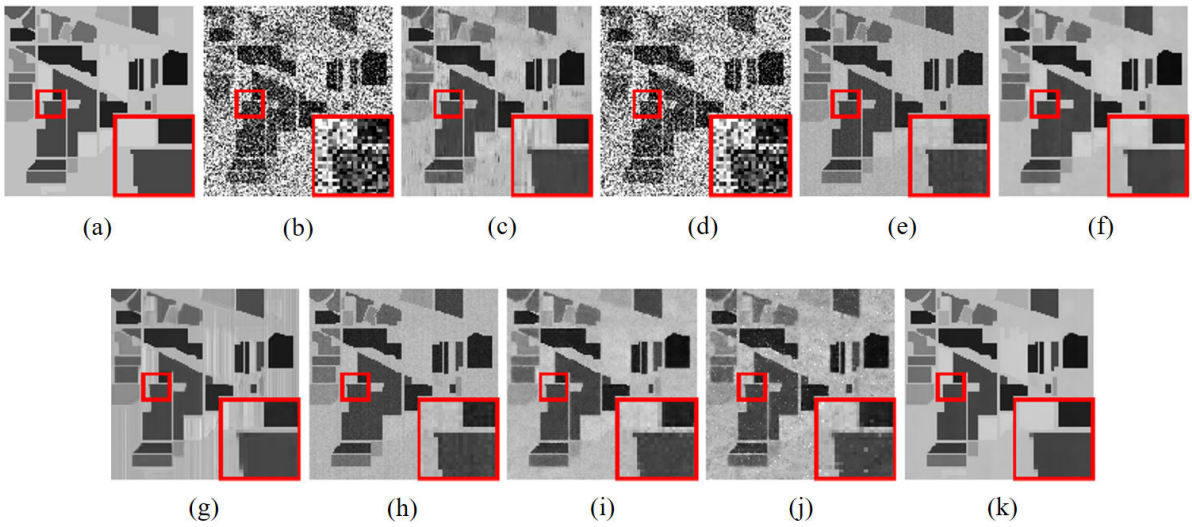


Fig. 3. Image of band 72 in the Indian Pines dataset before and after denoising by different methods. (a) Original data. (b) Noisy data. The restored results using (c) BM4D, (d) LRTV, (e) NMoG, (f) LRTDTV, (g) NGmeet, (h) NMoG-Tucker, (i) QRNN3D, (j) SQAD, and (k) WNLRATV.

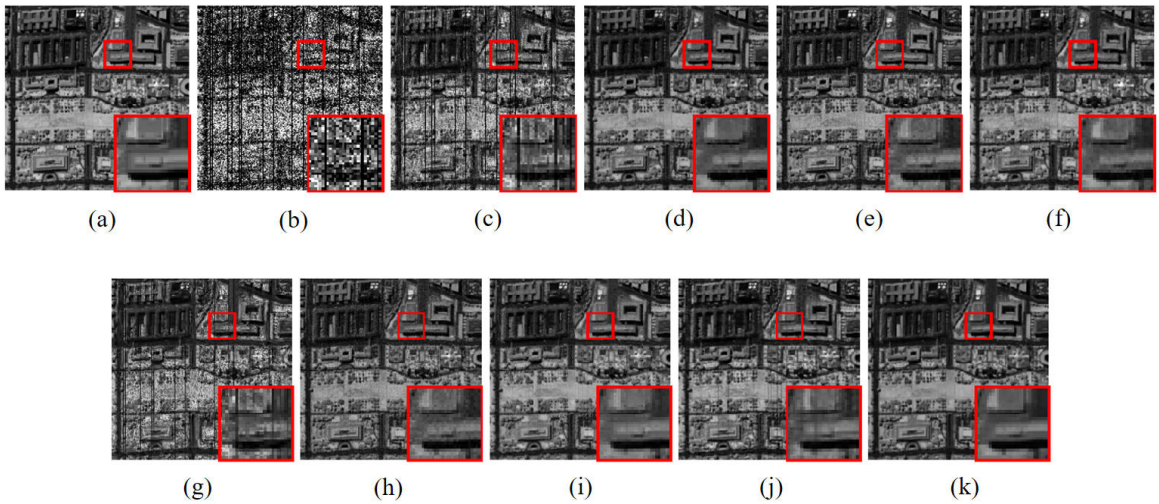


Fig. 4. Image of band 83 in the Washington DC Mall dataset before and after denoising by different methods. (a) Original data. (b) Noisy data. The restored results using (c) BM4D, (d) LRTV, (e) NMoG, (f) LRTDTV, (g) NGmeet, (h) NMoG-Tucker, (i) QRNN3D, (j) SQAD, and (k) WNLRATV.

3) Statistical Analysis: We also provide the statistical evidence of obtained quantitative results of different denoising methods using Friedman's test, to examine any significant variances between methods. Friedman's statistic is calculated as

$$\chi^2 = \frac{12N}{k(k+1)}\left(\sum_{i=1}^{k} r_i^2 - \frac{k(k+1)^2}{4}\right) \tag{41}$$

where N and k are the number of datasets and compared methods, respectively, and r_i represents the average ranking of each method over datasets. The statistic χ^2 is distributed according to chi-square with $k-1$ degrees of freedom. We record the average ranking of algorithms based on MPSNR as shown in Table III. The comparison was performed at a 0.05 level of significance. The test statistic of denoising results on Indian Pines is $\chi^2 = 31.04$, and the corresponding p-value is $p = 1.38 \times 10^{-4}$, which indicates the performance of methods is not equal. Similarly, the test statistic of denoising results on Indian Pines is $\chi^2 = 34.88$, and the corresponding p-value is $p = 2.81 \times 10^{-5}$, which indicates the variance does exist between all comparison methods. Furthermore, our method ranks first in the two simulated experiments, which shows that our method outperforms other methods.

B. Real Data Experiments

In this section, we further evaluate the performance of our method on two real HSI datasets, that is, the HYDICE Urban dataset[3] and airborne visible/infrared imaging spectrometer (AVIRIS) hyperspectral sensor dataset Indian Pines.[4] The HYDICE Urban dataset is with the size $307 \times 307 \times 210$. After cropping the main part of the data, it was resized to $256 \times 256 \times 210$. The AVIRIS Indian Pines dataset is with the size $145 \times 145 \times 220$. In both datasets, some bands

[3]http://www.tec.army.mil/hypercube

[4]https://engineering.purdue.edu/ biehl/MultiSpec/hyperspectral.html

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TABLE III
AVERAGE RANKINGS OF ALL COMPETING METHODS BASED ON MPSNR

	BM4D	LRTV	NMoG	LRTDTV	NGmeet	NMoG-Tucker	QRNN	SQAD	WNLRATV
Indian Pines	6.4	3.8	5.8	2.2	4.2	5.6	6.8	9.0	1.2
Washington DCmall	9.0	6.0	3.2	3.8	5.0	2.6	6.2	8.0	1.2

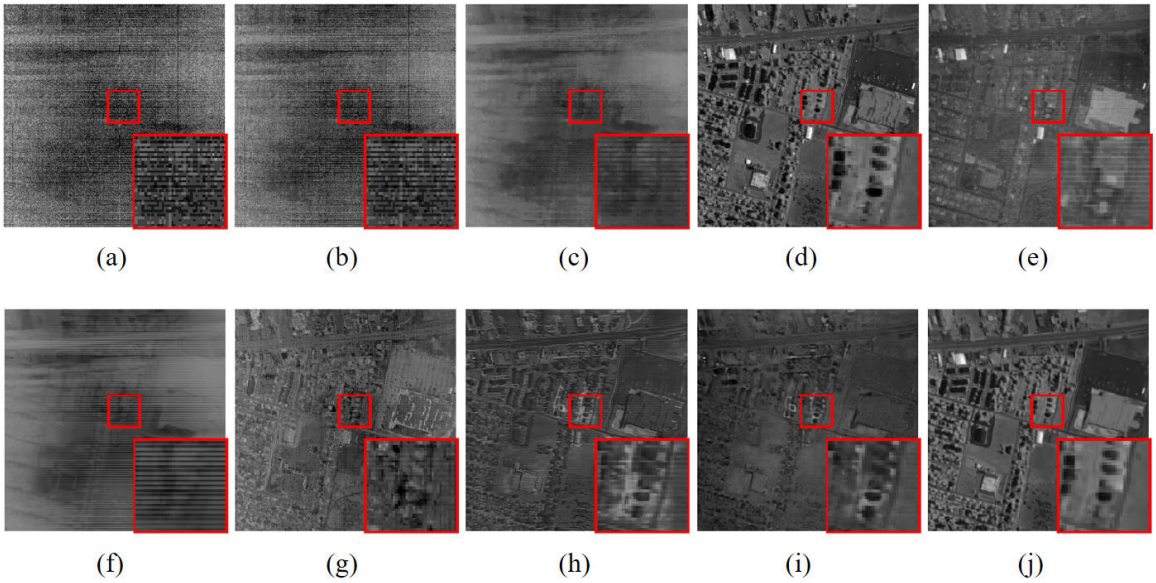


Fig. 5. Image of band 140 in the HYDICE Urban dataset before and after denoising by different methods. (a) Original data. The restored results using (b) BM4D, (c) LRTV, (d) NMoG, (e) LRTDTV, (f) NGmeet, (g) NMoG-Tucker, (h) QRNN3D, (i) SQAD, and (j) WNLRATV.

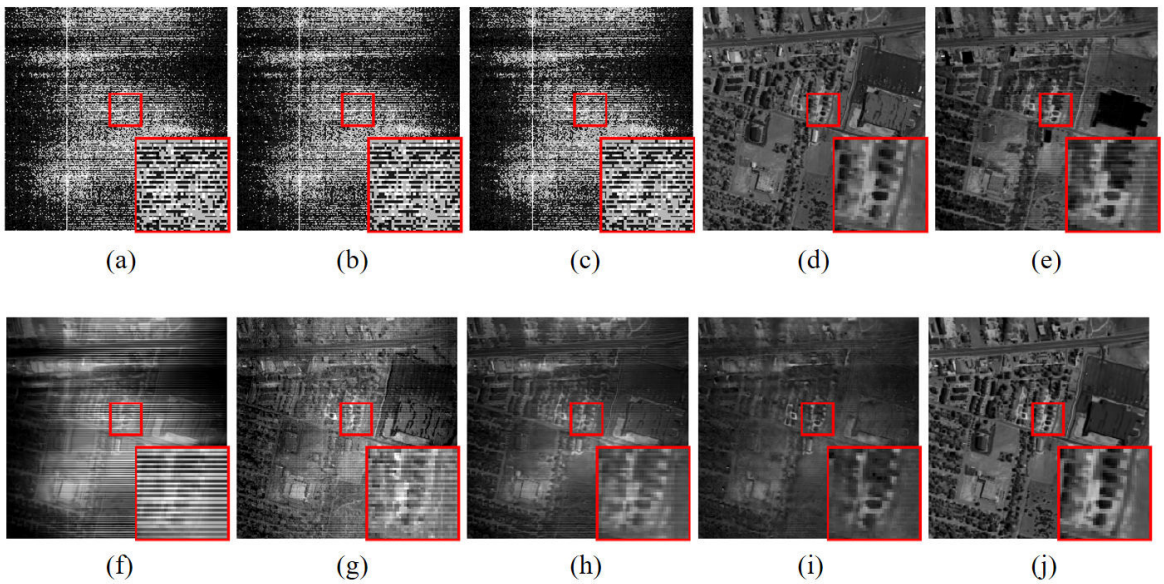


Fig. 6. Image of band 150 in the HYDICE Urban dataset before and after denoising by different methods. (a) Original data. The restored results using (b) BM4D, (c) LRTV, (d) NMoG, (e) LRTDTV, (f) NGmeet, (g) NMoG-Tucker, (h) QRNN3D, (i) SQAD, and (j) WNLRATV.

are heavily polluted by Gaussian noise, stripes, deadlines, the atmosphere, water absorption, or other unknown noise. We keep all of the data without removing any bands to test the efficacy of the proposed methods. The intensity range of the input HSI is scaled to $[0, 1]$. The competing methods are the same as the simulation experiment. As for the parameter setting, the rank of the proposed method is set as 6, which is the same as the NMoG method. For other competing methods, the parameters are set as suggested.

Figs. 5 and 6 demonstrate the restoration results of bands 140 and 150 in the HYDICE Urban dataset, respectively. It can be seen that the original bands 140 and 150 are overwhelmed by noise. BM4D method performs poorly in such complex noise cases due to the unreasonable i.i.d. Gaussian noise assumption. LRTV, LRTDTV, NGmeet, and NMoG-Tucker methods can remove part of the noise, while visible noise remains in the denoised results. The deep-learning denoising methods perform better in removing noises than these

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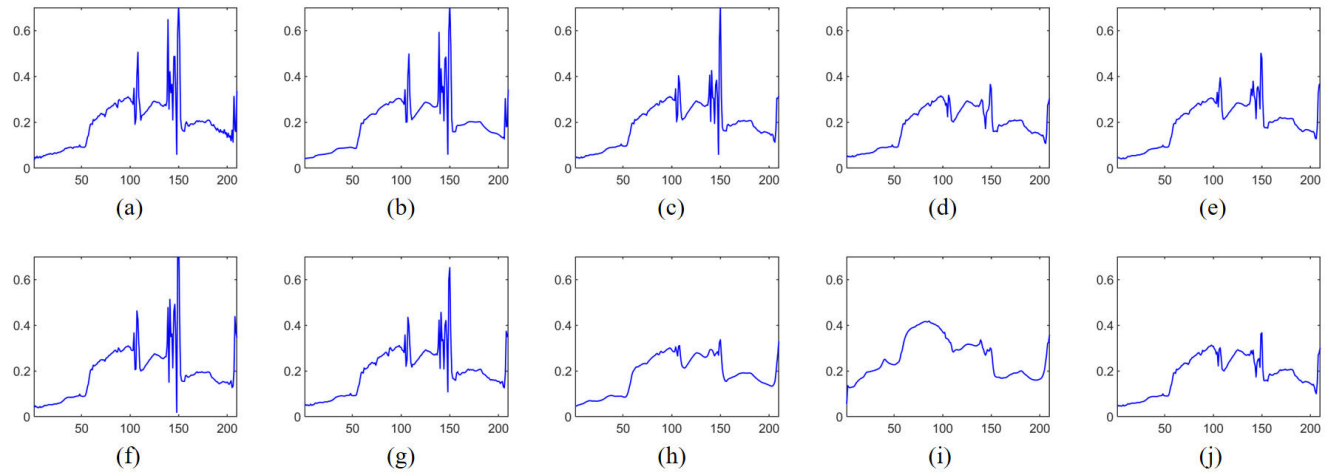


Fig. 7. Spectral signatures of point (185, 132) in the HYDICE Urban dataset before and after denoising by different methods. (a) Original data. The restored results using (b) BM4D, (c) LRTV, (d) NMoG, (e) LRTDTV, (f) NGmeet, (g) NMoG-Tucker, (h) QRNN3D, (i) SQAD, and (j) WNLRATV.

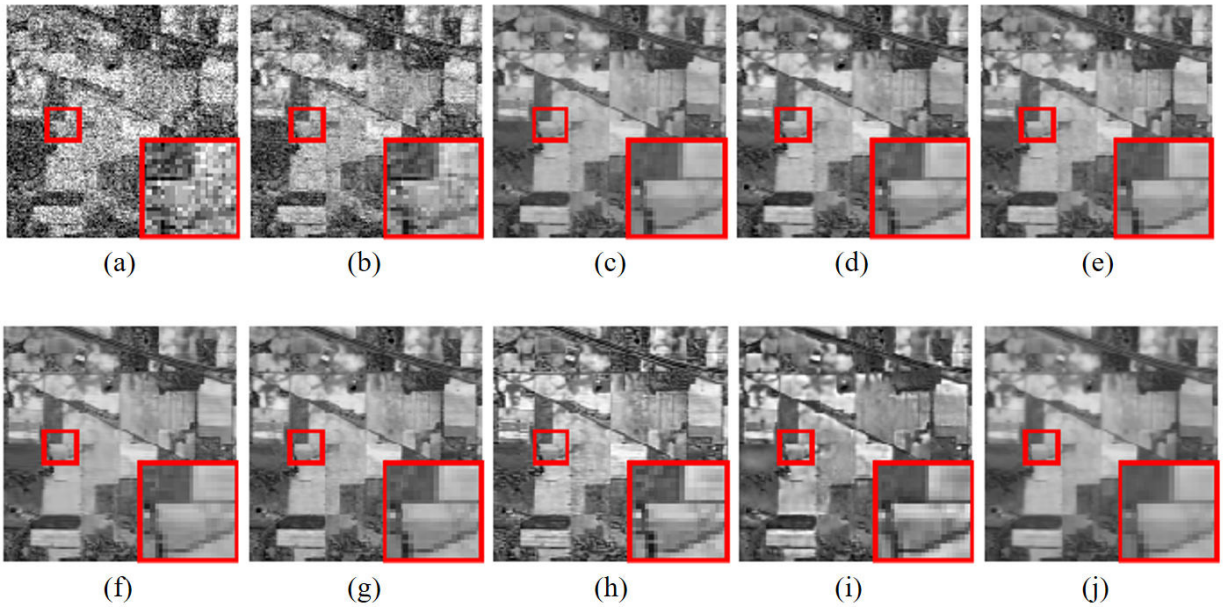


Fig. 8. Image of band 103 in the AVIRIS Indian Pines dataset before and after denoising by different methods. (a) Original data. The restored results using (b) BM4D, (c) LRTV, (d) NMoG, (e) LRTDTV, (f) NGmeet, (g) NMoG-Tucker, (h) QRNN3D, (i) SQAD, and (j) WNLRATV.

methods, but they bring data distortion in the denoised results. In contrast, NMoG and WNLRATV achieve the best denoising performance on this data. They visually not only remove the noise well, but also do not introduce image distortion.

To better demonstrate the denoising effect, the spectral signatures of the pixel located in (185, 132) before and after denoising were shown in Fig. 10. Due to the existence of noise, the curve shows rapid fluctuations as shown in Fig. 7(a). After denoising, the fluctuations are suppressed more or less. We can observe that the curve of QRNN3D and SQAD are oversmoothing, which is consistent with the data distortion as shown in Figs. 5 and 6. Significantly, the NMoG method and the proposed WNLRATV method have the most significant fluctuation suppression and keep the spectral reflectance characteristics well.

Figs. 8 and 9 demonstrate the restoration results of bands 103 and 220 in the AVIRIS Indian Pines dataset, respectively. It can be seen that the BM4D method fails to handle such severe degradation. Other competing methods perform much better, but still visually retain some noise in the denoised images. The proposed method yields a visually clean image, but the edge information is blurred. Furthermore, the spectral signatures of the pixel located in (105, 40) were shown in Fig. 7. We can observe that the curve rapidly fluctuates especially at the severely polluted bands as shown in Fig. 10(a). In the denoised results of BM4D, LRTV, NMoG, NGmeet, and LRTDTV, the curve becomes smoothing, but the fluctuations still exist at these bands which indicate the noise remaining. The curves of QRNN3D and SQAD are oversmooth, which indicates the information lost in the spectral domain. Although the curves of LRTDTV and WNLRATV

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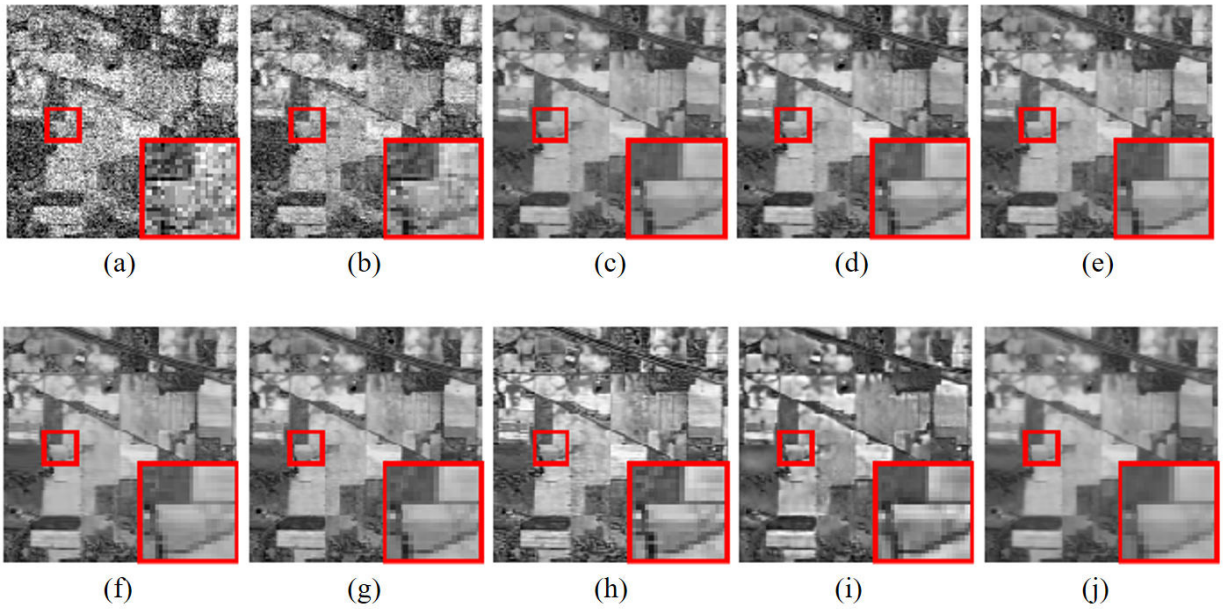


Fig. 9. Image of band 220 in the AVIRIS Indian Pines dataset before and after denoising by different methods. (a) Original data. The restored results using (b) BM4D, (c) LRTV, (d) NMoG, (e) LRTDTV, (f) NGmeet, (g) NMoG-Tucker, (h) QRNN3D, (i) SQAD, and (j) WNLRATV.

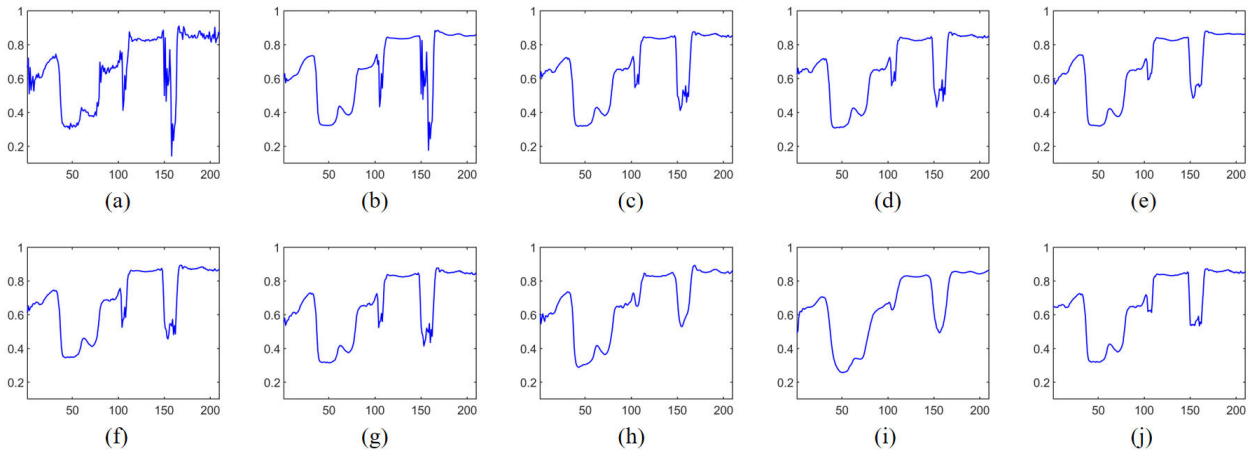


Fig. 10. Spectral signatures of point $(105, 40)$ in the AVIRIS Indian Pines dataset before and after denoising by different methods. (a) Original data. The restored results using (b) BM4D, (c) LRTV, (d) NMoG, (e) LRTDTV, (f) NGmeet, (g) NMoG-Tucker, (h) QRNN3D, (i) SQAD, and (j) WNLRATV.

are also over-smoothing, it retains more spectral reflectance characteristics compared to deep-learning denoising methods.

To further investigate the impact of denoising methods on subsequent tasks, we conduct a supervised classification of AVIRIS Indian Pines before and after denoised by the support vector machine (SVM) classification method. There are 16 classes, and the number of samples in each class ranges from 20 to 2455 in the ground-truth classification result. We randomly choose ten samples for each class as training data and use the remaining samples in each class for test. The classification results are quantitatively evaluated by the overall accuracy (OA). The classification experiment runs with ten random training datasets, and the best result was shown in Fig. 11. It can be seen that the OA of the original data is rather low, and all the denoising methods can improve the classification result more or less since the noise is suppressed. Among all denoising methods, our method leads to the highest OA value, demonstrating its superiority in benefiting the classification task.

C. Parameter Setting

In the WNLRATV method, there exists several parameters need to be set manually as shown in (27). Specifically, in the subproblem of updating $\mathcal{U}$ and V, the key factor that affects the solution is the ratio between ρ_1 and $\mathcal{W}$, denoting $\alpha = \rho_1/\bar{\mathcal{W}}$. The smaller the value of α, the higher the significance of observed data $\mathcal{Y}$ when updating $\mathcal{U}$ and V. In the subproblem of updating $\mathcal{C}$, the ratio between ρ_2 and ρ_1 directly affects the smoothness of $\mathcal{C}$. We denote $\beta = \rho_2/\rho_1$, The larger the value of β, the more smooth of $\mathcal{C}$. Similarly, in the subproblem of updating $\mathcal{A}$, the key factor is the ratio between λ_1 and ρ_2 that decides the sparsity of $\mathcal{A}$. We denote $\kappa = \lambda_1/\rho_2$, the larger the value of κ, the more sparsity of $\mathcal{A}$. Therefore, the parameters setting problem is transformed to

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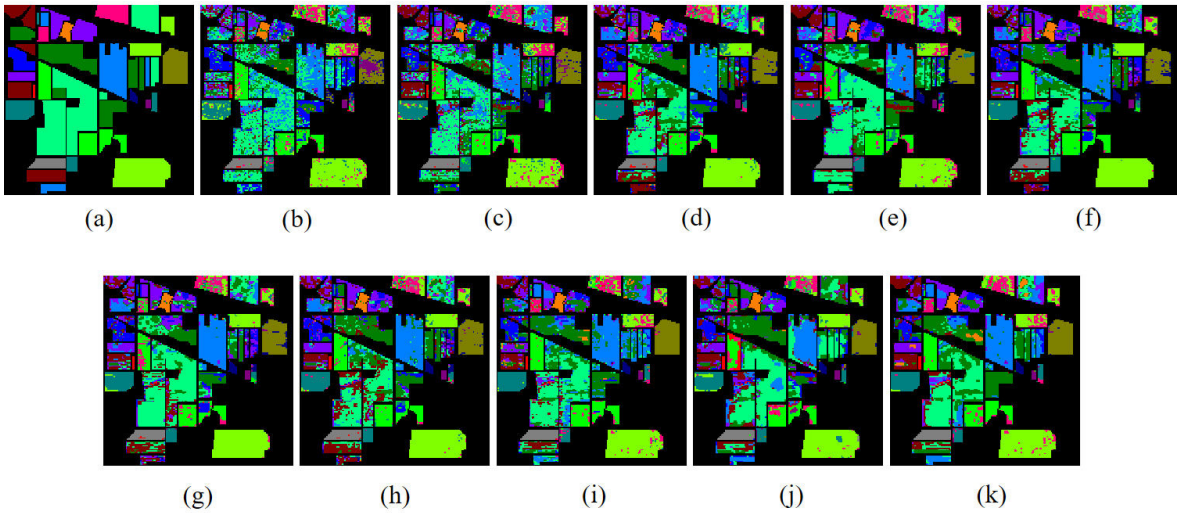


Fig. 11. Classification results by performing SVM on the AVIRIS Indian Pines dataset before and after denoising, and the corresponding OA values are reported. (a) Ground-truth classification. (b) Classification result on original data (OA = 49.12%). The classification results on restored data using (c) BM4D (OA = 54.30%), (d) LRTV (OA = 66.17%), (e) NMoG (OA = 66.90%), (f) LRTDTV (OA = 68.55%), (g) NGmeet (OA = 68.35%), (h) NMoG-Tucker (OA = 69.33%), (i) QRNN3D (OA = 66.73%), (j) SQAD (OA = 61.74%), and (k) WNLRATV (OA = 71.19%).

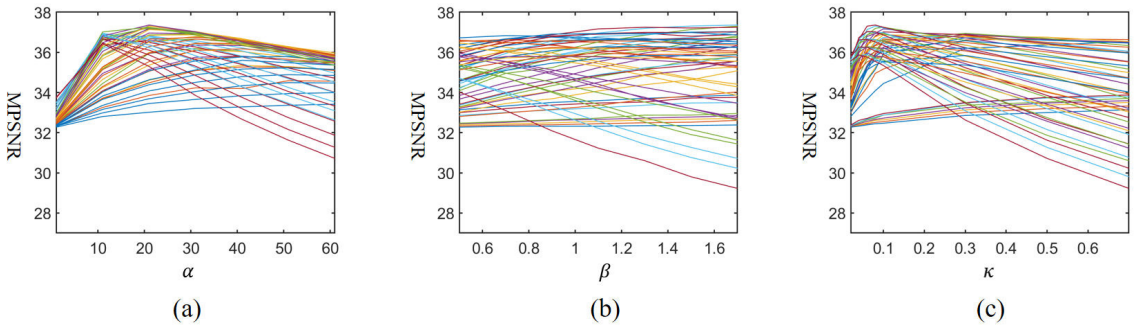


Fig. 12. (a) MPSNR values with different α setting; each line corresponds to a fixed set of (β, κ). (b) MPSNR values with different β setting; each line corresponds to a fixed set of (α, κ). (c) MPSNR values with different κ setting; each line corresponds to a fixed set of (α, β).

control the three ratios α, β, and κ. We can tune the ratios based on noise intensity, data sparsity, and smoothness.

Here, we also provide the parameter sensitivity analysis based on the synthetic Indian Pines dataset in the mixture noise case. We record the MPSNR measurement of denoised data under a set of parameter setting, that is, α varied in the set $\{1, 11, 21, 31, 41, 51, 61\}$, β varied in the set $\{0.5, 0.7, 0.9, 1.1, 1.3, 1.5, 1.7\}$, and κ varied in the set $\{0.02, 0.04, 0.06, 0.08, 0.1, 0.3, 0.5, 0.7\}$. Fig. 12(a) presents the MPSNR values with different α setting under a series of fixed (β, κ). We can see that the value of α has a significant impact on MPSNR. However, under most of (β, κ) combinations, the change in MPSNR is relatively stable when α is within the range $[10, 30]$, so we just need to fine-tune the value of α in this range. Fig. 12(b) shows the MPSNR values with different β setting under a series of fixed (α, κ). It can be clearly observed that MPSNR changes very slowly, indicating that the denoising result is insensitive to the value of β, and thus it can be fixed to 1 in all experiments. Fig. 12(c) presents the MPSNR values changing with the value of κ under a group of fixed (α, β). We can see that when the value of κ is too large or too small, the value of MPSNR will decrease, which means that the denoising effect will become worse. The optimal κ depends on the characteristic of the data structure. Specifically, when the data texture structure and spectral signature are simple, choose a large κ, otherwise a small κ is better.

VI. Conclusion

In this article, we propose a novel HSI denoising method by fully considering the complex noise structure and the intrinsic HSI prior. First, the non-i.i.d. noise modeling is utilized to model the HSI noise, which can adaptively learn the weighted loss function from data. Then, to encode the spatial–spectral smoothness prior, we propose a new ASSTV with the edge-preserving property. Finally, the nonlocal low-rank models are applied to encode the spatial and spectral correlation and the nonlocal similarity. The effective ADMM algorithm is developed to solve the proposed model. Experiments on simulated data and real data show that our method can achieve comparable results with state-of-the-art methods.

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Yang Chen received the Ph.D. degree from Xi'an Jiaotong University, Xi'an, China, in 2020.

From 2017 to 2019, she was a Visiting Scholar with the University of Illinois at Urbana-Champaign, Urbana, IL, USA. She is currently an Assistant Professor with the School of Mathematics and Statistics, Shaanxi Normal University, Xi'an. Her research interests include model-based deep learning and image processing.



Wenfei Cao received the Ph.D. degree in computer science from Xi'an Jiaotong University, Xi'an, China, in 2015.

He is currently an Associate Professor with the School of Mathematics and Statistics, Shaanxi Normal University, Xi'an. His research interests include matrix/tensor estimation, deep learning, and image processing.



Li Pang is currently pursuing the bachelor's degree with the School of Automation, Xi'an Jiaotong University, Xi'an, China.

His research interests include statistical modeling and image processing.



Xiangyong Cao received the B.Sc. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 2012 and 2018, respectively.

From 2016 to 2017, he was a Visiting Scholar with Columbia University, New York, NY, USA. He is currently an Associate Professor with the School of Computer Science and Technology, Xi'an Jiaotong University. He also works at the Ministry of Education Key Laboratory For Intelligent Networks and Network Security, Xi'an, China. His research interests include statistical modeling and image processing.

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